

and compute a present value of approximately 5.367%, of the notional, or around \$5,367,000.<sup>28</sup>

So how well did our handcheck do? The actual value of the swap (obtained by modeling the cash flows) is \$5,387,192, so our handcheck is off by around \$20,000. Is that a lot? Well, yes, \$20,000 is a lot of money, but in the context of this trade, it's pretty small. Let's look at it this way: if we go back to Table 3.9, we see that the PV Coupon 01 in a \$100mm 4-year swap with this swap curve is \$35,210. Thus, our handcheck gets us to within \$20,000/\$35,210  $\approx$  0.6 of a bp.<sup>29</sup> Not bad. Of course it's not exact, but we are under a bp away (in terms of bps running) from the actual value. So since the whole purpose of the handcheck is to just make sure we are in the right ballpark, we should feel pretty good about this.<sup>30</sup>

Why did we choose to set  $i = 5.47\%$  when valuing the annuity? This is a very reasonable question, and the answer is there is really no good answer. From the very get-go, it is inappropriate to value the annuity using a single yield. We should value cash flows using the appropriate zero coupon rates, but that would take too long: once again, the purpose of the handcheck is to do a back-of-the-envelope calculation with a few keystrokes.<sup>31</sup> So we are already making a fundamental error by discounting using a yield, so we shouldn't get hung up on what is the "right" yield to use to discount. It is wrong to discount using a yield in the first place, so there is no right yield

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<sup>28</sup>Be sure to set your financial calculator to assume that these annuity payments are received at the end of each year, not the beginning of each year. The first payment in the remaining 4-year swap is not today but one year from today, and the last payment is four years from today.

<sup>29</sup>Yet another way to handcheck this number is to take the PV Coupon 01 for a 4-year swap in Table 3.9 and multiply it by 153, the number of bps in the annuity. This gives us  $\$35,210 \times 153 = \$5,387,130$ , which is very close to the model-based number. The only reason that this number does not tie out exactly with our model is because the PV Coupon 01 numbers presented in Table 3.9 are rounded to the nearest dollar, introducing a small rounding error. In general, we will not rely on this type of handcheck because it requires knowing what the PV Coupon 01 is, and we want a handcheck methodology that does not rely upon use of a swap model at all.

<sup>30</sup>What is a reasonable margin of error on a handcheck? That is a matter of personal opinion, but many in the industry start to get uncomfortable when off by more than a bp running.

<sup>31</sup>Suppose we do use the zero coupon rates to exactly value the annuity. In that case we get the value of the swap is

$$\frac{\$1,530,000}{1.05} + \frac{\$1,530,000}{(1.05113)^2} + \frac{\$1,530,000}{(1.05312)^3} + \frac{\$1,530,000}{(1.05494)^4} = \$5,387,192,$$

the exact answer from our model. But that took way too much math to count for a quick handcheck. By the way, please note that the zero coupon rates shown here are rounded and thus won't exactly match the actual unwind value. If the exact zero coupon rates are used, the values will match exactly.

to use anyway. We should just use something reasonable. We need to plug in something for  $i$ , which is used to discount four annual cash flows, so it seems pretty reasonable to use the 4-year swap rate. In fact, you see that the 4-year swap rate of 5.47% looks like it lies somewhere in middle of the four zero coupon rates (5%, 5.113%, 5.312%, and 5.494%) one would use if being precise, so that should give us some comfort. But let's not worry too much about what yield we throw in there for discounting purposes, as long as we choose something sensible. If we set  $i = 20\%$ , that wouldn't seem too sensible in the swap curve environment we were given, and we would probably have trouble convincing people it was a good choice.  $\square$

## Example 2

Let's now look at the market information for swaps on the composite page shown in Table 1.1, where the standard convention is fixed payments are made semi bond and the floating side pays 3-month LIBOR quarterly money. Suppose that exactly three years ago a client entered into a \$100mm 8-year swap with a dealer in which the client receives 4.647% semi bond and pays 3-month LIBOR quarterly money. The client is interested in knowing the mark-to-market value of the swap. Let's assume that mark-to-markets are done by valuing swaps at mid-market swap rates prevailing in the market.

## Handcheck for Example 2

Since our swap model is only set up for annual coupons, we are not going to be able to model up this swap. But we can use a handcheck approach to approximate the current mark-to-market. The first three years of the swap have come and gone and have no bearing on the current valuation of the swap. The client is currently receiving 4.647% when the mid-market 5-year swap rate 1.627% (the average of the bid and offer side swap rates). The approach to handchecking this swap is very similar to what we did in Example 1, but we have to be very careful here with the frequency of payments, which in this example are semiannual on the fixed side. The strategy for valuing this swap is shown in Figure 3.12.

Once again we can add in the bottom box because it has zero value; 1.627% is the mid-market 5-year swap rate, and thus paying or receiving this rate in a 5-year swap is worth nothing. Note that the LIBOR legs in the bottom two boxes, which in this case are paid quarterly, still cancel each other out exactly. And so we are left with the client receiving an annuity of  $4.647\% - 1.627\% = 3.02\%$  that is paid semi bond. The swap is in-the-money to the client, since the client is receiving a fixed rate above the 5-year swap

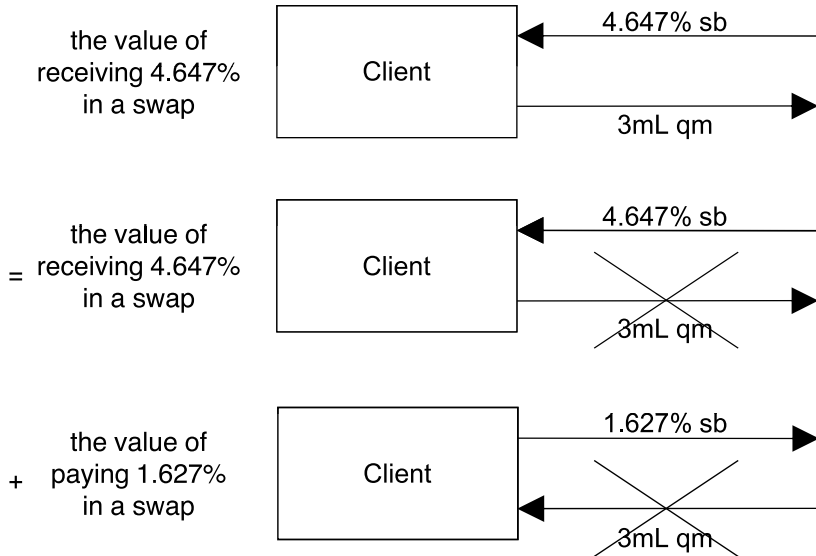


Figure 3.12: Looking at the Value of the 4.647% Swap with Standard Payment Conventions

rate. To value the swap using a handcheck approach, put

$$\text{PMT} = (4.647\% - 1.627\%)/2 = 1.51\%$$

$$n = 5 \times 2 = 10$$

$$i = 1.627\%/2 = 0.8135\%$$

in a financial calculator, and we get a present value of around 14.446% of the notional, or around \$14,446,000. The important thing to note in this example is that we had to look at the smallest frequency in which fixed payments were made. In this case the swap makes payments semiannually on the fixed side, and it is important to account for the timing of these cash flows when discounting back to find the present value of the annuity.<sup>32</sup> □

<sup>32</sup>If we had forgotten to account for the fact that the fixed side pays semiannually, our handcheck on this would have (inaccurately) been found by putting  $\text{PMT} = 4.647\% - 1.627\% = 3.02\%$ ,  $n = 5$ , and  $i = 1.627\%$  and getting  $\text{PV} = 14.390\%$  of the notional, or \$14,390,000, around \$56,000 off from the (more appropriate) handcheck amount of \$14,446,000. Although handchecks are approximate, not accounting for the correct frequency of payments would have introduced an unnecessary rounding error.

### 3.3.2 Unwinds

An *unwind* of an existing swap is an agreement between the two counterparties to have no further obligations to one another as per the original terms of the swap. An *unwind fee* will be made from one party to the other reflecting the current market value of the swap, so as to compensate the party that is in-the-money to forgo the continuation of the swap. This fee is typically paid  $T + 2$  days after the trade has been unwound.

#### Example 3

Let's continue with Example 2. Suppose that based upon the mark-to-market we just calculated, the client would like to unwind the trade. Recall that in the original trade, the client is receiving fixed. Now, in order to unwind this trade, the dealer is implicitly being asked to receive fixed from the client to reverse the original trade that was put on. That is, hypothetically, if the client were at this moment in time to pay fixed in a 5-year swap at a rate of 4.647% semi bond versus threes, this would exactly negate the position the client currently has on. And presumably the dealer would be willing to pay something to the client for this, since that rate is well above the current offer side 5-year swap rate of 1.647%.

The dealer receives fixed at the offer side swap rate, which in this case is 1.647%, and the amount of the unwind fee is determined relative to this offer side swap rate. Since the dealer would normally receive 1.647% in a 5-year swap executed right now but will instead receive a higher rate of 4.647%, this implies that the dealer will pay an unwind fee to the client for terminating the swap early.

#### Handcheck for Example 3

Since this example also has semiannual payments on the fixed side and thus does not fit into the paradigm of our model, we will have to come up with a handcheck for the amount of the unwind fee. The logic of computing the amount of the unwind fee is shown in Figure 3.13. When it comes to unwinding the swap, the dealer is essentially being asked to receive 4.647% versus paying 3-month LIBOR. This is not literally the case, but that is the economic impact of unwinding the swap. As Figure 3.13 demonstrates, the unwind of the swap (represented in the top transaction) can be broken up into the dealer receiving the offer side swap rate of 1.647% semi bond versus 3-month LIBOR (the middle transaction) plus the dealer receiving an annuity of what is left over:  $4.647\% - 1.647\%$  semi bond (the bottom transaction). Since the dealer is receiving the offer side swap rate in the middle

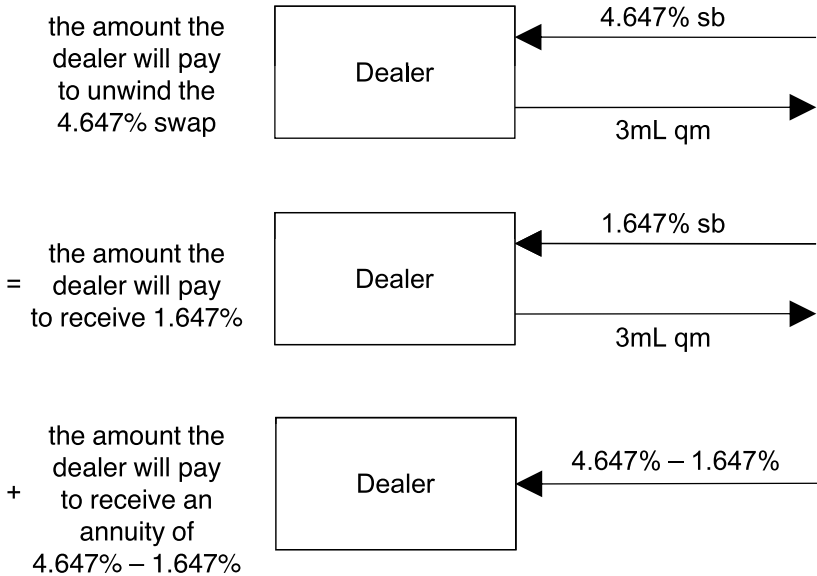


Figure 3.13: Determining the Unwind Fee in a Swap in Which the Client Is Currently Receiving 4.647%

box, there will be no upfront fee associated with this piece. To determine the amount the dealer will pay for the annuity in the third box, we put

$$\begin{aligned} \text{PMT} &= (4.647\% - 1.647\%)/2 = 1.50\% \\ n &= 5 \times 2 = 10 \\ i &= 1.647\%/2 = 0.8235\% \end{aligned}$$

in a financial calculator, and we solve for  $PV = 14.342\%$  of the notional, or around \$14,342,000. Note that the amount that the dealer is willing to pay to unwind the trade, \$14,342,000, is less than the mark-to-market of the trade, \$14,446,000, determined in Example 2. The difference represents the bid/offer that the dealer is making on the unwind.  $\square$

### 3.3.3 Assignments

In Examples 2 and 3 we considered a client who had originally entered into an 8-year swap with a particular dealer, which we will denote by Dealer 1. Under normal market conditions, this does not mean, however, that the client is beholden to Dealer 1 for purposes of unwinding the trade. If he wishes,

the client can do an *assignment* of the trade to another dealer. That is, the client can always contact another dealer, which we will denote by Dealer 2, to compete on the unwind. Dealer 2 will be asked what upfront fee it is willing to pay to the client in order to receive 4.647% on a \$100mm swap for five years. If Dealer 2 is willing to receive fixed in a 5-year swap at a more aggressive rate than Dealer 1, then Dealer 2 will be able to provide a larger upfront amount to the client.<sup>33</sup> In this case, instead of unwinding the trade, the trade will be *assigned*. Dealer 2 will pay the upfront fee to the client and will now receive 4.647% on a \$100mm swap for the next five years, facing Dealer 1. In the case of this assignment, the client is known as the *assignor*, Dealer 2 is known as the *assignee*, and Dealer 1 is known as the *remaining party*.

There's one catch here. The client can't assign the swap to Dealer 2 unless Dealer 1 has given its permission to assign the swap (and presumably when the client contacted Dealer 2, it checked with Dealer 2 that it would be willing to face Dealer 1 in the swap). Remember, a swap is a contract between two counterparties and carries with it counterparty risk. It could be the case that Dealer 1 and Dealer 2 don't even have an ISDA Master Agreement in place with one another to do swaps. This is rather unlikely, however, as most dealers have documentation in place with all other dealers so that they can do such trades on assignment, or face other dealers on trades done through the brokers market. It is conceivable, however, particularly during times of financial stress, that one dealer may refuse to accept another dealer on assignment. The reputational effects of "turning down a name," both for the party who refuses but especially for the party who is refused, are potentially enormous.

### 3.3.4 Forward Starting Swaps

A *forward starting swap* is a swap that has a settlement date beyond the standard  $T + 2$  New York and London business days. To take a particular example, consider a 3-year swap, one-year forward (also known as a "1x4", which is pronounced "1 by 4").<sup>34</sup> This swap has an effective date that is in

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<sup>33</sup>Suppose, for example, that Dealer 2 was willing to receive in a 5-year swap at a rate of 1.637%. This is 1 bp more aggressive than a 5-year offer side rate of 1.647% showed by Dealer 1. In the context of Figure 3.13, this means that Dealer 2 will value the existing swap as an annuity worth  $4.647\% - 1.637\% = 301$  bps to the client, as compared to the 300 bp annuity that Dealer 1 is valuing it at.

<sup>34</sup>Some also refer to this as a "1 into 3," although this terminology lends itself to possible confusion since swaptions, which will be discussed in Chapter 5, employ the " $x$  into  $y$ " terminology as well.

| Time | zc     | Forward    | Fixed       | PV Fixed   | Float       | PV Float     |
|------|--------|------------|-------------|------------|-------------|--------------|
| 1    | 5.000% | 5.000%     | 0           | 0          | 0           | 0            |
| 2    | 5.113% | 5.226%     | 5,644,262   | 5,108,529  | (5,225,748) | (4,729,739)  |
| 3    | 5.312% | 5.712%     | 5,644,262   | 4,832,487  | (5,712,202) | (4,890,656)  |
| 4    | 5.494% | 6.041%     | 5,644,262   | 4,557,172  | (6,041,364) | (4,877,792)  |
|      |        |            |             | 14,498,187 |             | (14,498,187) |
|      |        | Fixed rate | 5.644%      | NPV        | 0           |              |
|      |        | Notional   | 100,000,000 |            |             |              |

Figure 3.14: Pricing Up a 1x4 Forward Starting Swap

one year.<sup>35</sup> Assuming the swap pays fixed semi bond versus 3-month LIBOR quarterly money, then the first floating payment would be in 1.25 years from today, the first fixed payment would be in 1.5 years from today, and final payments on both sides would be made four years from today.

The fixed coupon in a forward starting swap is set such that the swap is an  $NPV = 0$  transaction. If we think about pricing swaps using Eurodollar futures, recall that a spot starting swap is priced off deposit rates (e.g., today's LIBOR setting) and a sequence of Eurodollar futures including the one that is next to settle serially. A forward starting swap may very well be priced off of a deferred sequence of Eurodollar futures, omitting one or more contracts that are next to settle serially. It is possible, however, to also consider pricing a forward starting swap as a function of observable spot starting swap rates, as we will soon see.

### Example

Consider again our stylized example, and let's price up a 1x4 (i.e., a 3-year swap, one year forward) in which payments on both sides are made annual bond. Figure 3.14 shows how we price up the 1x4. All of our familiar zero coupon rates and forwards are there in the model. However, notice that there are no cash flows that occur at the end of the first year. The 1x4 is effective in one year, and thus its first cash flows are at the end of the second year. The first forward (which is actually today's spot LIBOR) is not employed

<sup>35</sup>To be completely precise, the exact effective date would be one year after the standard  $T + 2$  settlement date of spot starting swaps.

in the model to form the floating cash flow at the end of the first year, since there is no cash flow in that year. The remaining three forwards are used to determine the market's expectation of the floating flows in the swap over its three-year life. The fixed rate that we solve for, 5.644%, is such that the present value of the three fixed flows equals the present value of the three floating flows.<sup>36</sup> □

## Hedging Forward Starting Swaps

We've previously talked about how a dealer would hedge a spot starting swap with either Treasuries or Eurodollar futures. A 1x4 has a final maturity that is four years from today, but does that mean that the dealer should hedge this forward starting swap in the same way it would hedge a spot starting swap?

The answer is no. Since there are no cash flows at all in the first year, any hedge that had interest rate exposure in the first year would not be ideal. One good hedge would be to trade in a deferred strip of Eurodollar futures. Yet another way the dealer could hedge out his position in a forward starting swap is by trading in two different spot starting swaps. Let's continue with our example in which the dealer is receiving fixed in a 1x4. As we will see, the dealer can hedge himself by simultaneously paying fixed in a 4-year spot starting swap and receiving fixed on the same notional of a 1-year spot starting swap. Intuitively, by paying fixed for four years, but then receiving fixed for one year, the dealer will negate the position in the first year but maintain exposure in the back three years. This principle is suggested in Figure 3.15. In fact, this figure will be the basis for the handcheck of a forward starting swap rate that we will now develop.

## Handcheck

It used to be that there was very little information about forward starting swap rates that was made available on any pricing screens. People would rely on their own swap models to see where a particular forward starting swap rate should be. Over the years, more information about forward starting swap rates has become widely available. Many traders will keep an eye on a forward starting swap matrix, such as the one in Table 3.10, looking for opportunities to put on interesting trades (some of which we will discuss later in this chapter and others of which we will discuss in Chapter 8). In this

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<sup>36</sup>Note that in Figure 3.14, we are using the three forward rates  $f(1,2)$ ,  $f(2,3)$  and  $f(3,4)$  for purposes of computing the market's expected nominal cash flows in the floating leg of the swap. And we use  $zc(2)$ ,  $zc(3)$ , and  $zc(4)$ , respectively, to form discount factors in order to compute the present value of these nominal cash flows.



THE PRICING OF SWAPS

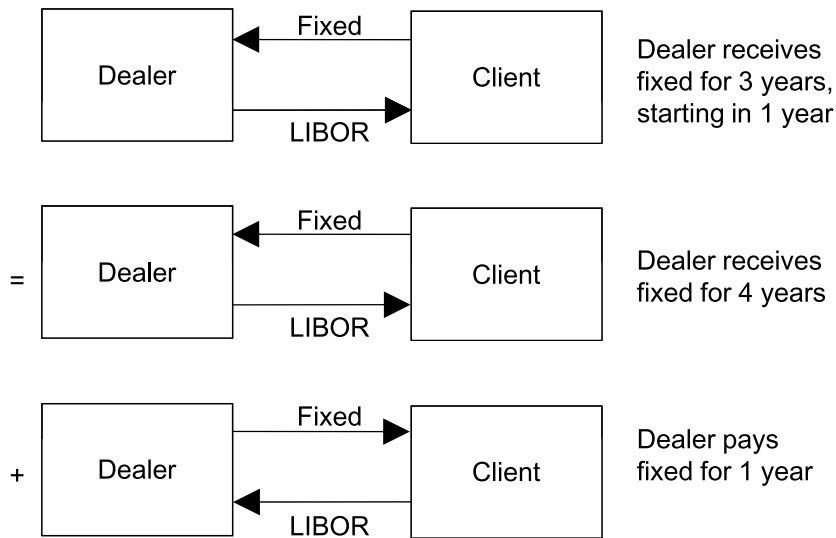


Figure 3.15: Replicating a 1x4 Forward Starting Swap

|         |     | Swap |      |      |      |      |      |      |      |      |
|---------|-----|------|------|------|------|------|------|------|------|------|
|         |     | 1    | 2    | 3    | 5    | 7    | 10   | 15   | 20   | 30   |
| Forward | 0m  | 1.62 | 2.25 | 2.77 | 3.48 | 4.00 | 4.49 | 5.00 | 5.22 | 5.31 |
|         | 3m  | 1.84 | 2.52 | 3.00 | 3.67 | 4.16 | 4.62 | 5.10 | 5.30 | 5.37 |
|         | 6m  | 2.15 | 2.80 | 3.24 | 3.86 | 4.31 | 4.74 | 5.20 | 5.38 | 5.44 |
|         | 1y  | 2.88 | 3.36 | 3.70 | 4.24 | 4.62 | 4.99 | 5.39 | 5.53 | 5.56 |
|         | 2y  | 3.87 | 4.14 | 4.39 | 4.81 | 5.08 | 5.38 | 5.69 | 5.77 | 5.74 |
|         | 3y  | 4.43 | 4.67 | 4.89 | 5.21 | 5.42 | 5.68 | 5.90 | 5.94 | 5.87 |
|         | 4y  | 4.91 | 5.14 | 5.30 | 5.52 | 5.70 | 5.92 | 6.07 | 6.07 | 5.96 |
|         | 5y  | 5.38 | 5.52 | 5.62 | 5.78 | 5.94 | 6.12 | 6.20 | 6.16 | 6.03 |
|         | 7y  | 5.83 | 5.91 | 5.99 | 6.14 | 6.28 | 6.36 | 6.34 | 6.25 | 6.07 |
|         | 10y | 6.33 | 6.42 | 6.49 | 6.58 | 6.59 | 6.53 | 6.39 | 6.26 | 6.04 |

Table 3.10: Forward Starting Swap Rate Matrix

matrix, the first row refers to spot starting swaps of a variety of maturities (i.e., the swap curve). Subsequent rows contain forward swap rates for a given delay that are derived from the given swap curve. For example, in this matrix, the 10-year swap rate is 4.49%, and the 10-year swap rate, 1 year forward (denoted 1x11) is 4.99%.

Despite the proliferation of data on forward starting swaps, an understanding of how the rates are derived from spot swap rates is crucial. We already looked at pricing forward starting swaps in our model, as shown in Figure 3.14, and we are now going to develop a handcheck for the computation of this rate. This handcheck methodology will be based upon the logic suggested in Figure 3.15. Let's look at this figure again. Suppose for the moment that the 4-year swap rate happened to equal the 1-year swap rate, and denote this rate by  $x$ . If these two rates are the same, then all cash flows in the bottom two swaps of this figure would exactly cancel out over the first year. That is, the two fixed rates would cancel each other out, and the two floating rates would cancel each other out for the first year. So we would have no cash flow in the first year, and in the back three years the dealer would be receiving fixed at a rate of  $x$  and paying LIBOR. But this would mean that the 1x4 swap rate is  $x$ .

In general, the 4-year swap rate will not equal the 1-year swap rate, and thus not all of the cash flows in the first year will exactly cancel each other out. But all is not lost; we can still proceed with this methodology as long as we are careful to "take" any cash flow that would occur in year one and "move" it to the back three years. When we say "move," we mean that we will take any cash flow that occurs in year one and spread it out over the life of the swap (accounting for its value appropriately, of course) in years two, three, and four. That is, given any cash flow at the end of year one, we could ask ourselves what annuity payment we could receive at the end of years two, three, and four that has the same present value. Based on this answer, we are then indifferent between receiving the cash flow at the end of the first year or the annuity over the back three years, and we can move the cash flow from year one into the back three years of the life of the swap.

### Example 1

Assume that the swap curve is as given in Table 3.5. Let's look at what happens if the dealer receives in four years at 5.47% and pays in one year at 5.00% (let's refer to this as the "replicating strategy" for the forward starting swap). The important thing to keep in mind is that this replicating strategy will imply what the fixed rate in the 1x4 swap must be. To see this, first note that the two LIBOR legs will exactly cancel out over the first year, but the dealer is left paying LIBOR at the end of years two, three, and four. This is good: in the forward starting swap the dealer also pays LIBOR in years two, three, and four. So by entering into the replicating strategy, we emulate the floating leg of the forward starting swap perfectly.

Let's now look at the fixed flows that result from the replicating strategy. Since the two fixed rates differ, this means that the dealer will receive a net amount of 47 bps (the difference in the two fixed rates) at the end of year one